Rectina, a dense ink reader for Herculaneum papyri trained from scratch on composite labels

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***Abstract*—Rectina is an ink reader for Herculaneum papyri: the villa hybrid 3D/2D network, trained from random weights on dense composite labels of eight scrolls, then trained further on larger tiles following the recipe of Reader v3 (30000 iterations at 256 px, 20000 at 384 px). Two checkpoints are released, m1 after the first stage and m2 after the two further stages. The differences from Reader v3 are the start from random weights and a per-patch depth warp with its normalisation guard in the first stage; there is one seed per checkpoint and no ablation, so no result is attributed to either. The rule for adopting m2 was fixed in advance: above Reader v3 on at least 6 of the eight held-out units with a median gain above 0.003. It is met (6 of 8, median +0.0137), but four of the six variants scored meet it too, and the held-out units are segments of scrolls used in training, so the rule is descriptive. On the six control units outside the training segments m2 is above Reader v3 on 5 of 6 (median +0.0040), and on the four with organisers' labels on 3 of 4 (median +0.0063): differences of a few thousandths, below what this work can resolve with one seed. Against Reader v2, m1 is above on 8 of 8 held-out units (median +0.0270) and on 4 of 6 controls (median +0.0104). Weights and model cards are published in the model repository [1]; the reader, the render recipe and the evidence files in the article folder [2].**

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I. Introduction

The organisers of the Vesuvius Challenge state the last of the three problems of reading a scroll as "detecting or segmenting the ink signal on, or near, the recovered writing surface" and add that "The subtle differences that distinguish ink from papyrus must be teased out of data like this" [3]. A detector that answers this must say, for each point of a segmented surface, whether it carries ink. The readers that exist for the scrolls of the open data [4] are Reader v2 [5] and Reader v3 [6].

The engine is not ours: the scan data and the labels of the organisers, the ink detection code of villa with its model, training and render tools [7], and the two readers with which we compare. The training in stages on larger tiles follows Reader v3, which in the words of its README is "Reader V3 is Reader v2 trained further on larger tiles, 30k iterations at 256 px and then 20k at 384 px", and for which "The architecture, the training data and the labels stayed the same" [6]. What is ours, as the files of this work show: the evaluation on units outside the training segments with its overlap check, the render recipe with its patch, the checks of the comparison and the chain that releases the checkpoints, and, as differences from Reader v3, a start from random weights (the first training step was continued in place from iteration 79001 of an earlier run of the same recipe) and a per patch depth warp with its normalisation guard in stage 1 in place of the flat random shift of the depth window that the configuration of Reader v3 names. We make no claim that the recipe or any of these differences explains a result: there is one training seed for each checkpoint and no ablation. Rectina is a dense ink reader, a three dimensional network that takes a stack of seventeen depth layers of a scroll volume and returns for every pixel of the stack surface the probability that it carries ink. Rectina is released in two checkpoints, m1 after the first training step and m2 after two more. This article gives the recipe of the training, the measurement, and the comparison of Rectina m1 with Reader v2 and Reader v3 and of Rectina m2 with Reader v3, on eight held-out units and on six control units, with the same scores on the same units.

The rule of the abstract and the controls are the results. Against Reader v2, Rectina m1 scores above on 8 of the eight held-out units (median gain +0.0270), on the six controls on 4 of 6 (median +0.0104) and on the four with organisers' labels on 3 of 4 (median +0.0104); of the six variants scored, six meet the rule against Reader v2 and four against Reader v3. Descriptively, over all fourteen units, Rectina m1 scores above Reader v2 on 12 of 14 with a median difference of +0.0200, and Rectina m2 scores above Reader v3 on 11 of 14 with a median difference of +0.0067; against Reader v3 the count of Rectina m1 is 9 of 14, with a median difference of +0.0071.

Not claimed: a new architecture (the model is the one of villa), a result on any scroll beyond the fourteen units, a calibrated probability, or a difference between two recipes. Each checkpoint comes from one training seed, the paired tests are not corrected for multiple comparisons, and the scores of different readers are not calibrated against each other. No earlier work of this series is cited, because none of its modules appears here.

II. Starting point

The starting point is the ink detection code of the villa repository, in the aligned21 hybrid 3D/2D configuration of villa commit 3ea17f54a9b3, whose base settings are listed in appendix A beside the settings we changed. The readers against which we compare are the published Reader v2 and Reader v3.

III. Surfaces and renders

The models read a stack of seventeen depth layers of the scroll volume and return one probability for every pixel of the surface of the stack. The units used to score them are cut

from the volumes on windows that were fixed from the labels before any map existed.[1]

IV. Labels and validation

The training labels are those of the label set `43965c9d120c`: 103 units in all, from eight scrolls (the scrolls of the per-scroll quotas of appendix A). The set is composite in that its units are of three kinds: 74 carry labels moved between scan resolutions and placed with a registered transformation and offset, 5 are native units whose shift was measured, and 24 are composite units on a canvas of 9.596 um. The fourteen units used to score the models here were kept out of every training set of these models.[2] Validation: for each training segment, whole 512 px tiles holding 5 percent of its supervised pixels (selection seed 20261017) are kept out of training.

The offset we found and reported in villa issue 2243 was corrected in the labels used here. Labels moved between scan resolutions were placed with that correction (7.2 to 7.4 px, registered per segment) on 74 of the 103 units, by measured shifts on 5 native units; the remaining 24 composite units were checked only by a search within 4 px.

The scan data were obtained from the Vesuvius Challenge data server (dl.ash2txt.org) under the data licence of the Vesuvius Challenge (LICENSE.txt).

V. Training recipe

The recipe is given as the stages in which the models were trained and the setting of each stage. A setting that is not repeated in a stage is the one of the stage before it. Stage 1 gives Rectina m1; stages 3 and 4 follow it and give Rectina m2; the stages carry the numbers of their configuration files (s1, s3, s4). Rectina m1 is the checkpoint taken at the fixed step 130001 (file ckpt_130001; embedded step counter 130000), and Rectina m2 is the final step of stage 4 (file ckpt_020000; embedded step counter 19999). Each value was read by a tool from the configuration file that was used, and the table of appendix A gives the source of each value, with file and line where the source is public code.

The composite labels are the set defined in section IV; we do not compare them with the labels of Reader v3.

Training runs in three steps on one label set, each step starting from the checkpoint of the step before; the patch grows from step to step and the batch shrinks after the first step. The checkpoint at the end of the first step is released as Rectina m1, and the checkpoint at the end of the third step as Rectina m2.

Stage 1 (s1), released as Rectina m1: 131000 iterations (taken at its fixed step, ckpt_130001), batch 64 patches per step with per scroll quotas (0139 12, 1667 8, Paris4 6, 0814 7, 0009B 8, 0343P 5, 0500P2 7, 0172 11), patches of 17x128x128 voxels (z, y, x), learning rate 0.01, optimizer sgd, warmup 1000 steps, validation every 1000 steps, depth warp augmentation with a maximum offset of 3 planes. The base settings are those of the aligned21 hybrid 3D/2D configuration of the villa ink detection code (optimizer sgd, learning rate 0.01, batch size 64, patch 17x128x128, precision fp16). The run was continued in place, with the full training state and the data sampler advanced, from the checkpoint at iteration 79001 of an earlier run of the same stage.[3]

Stage 3 (s3), the first of the two further stages: from the s1 checkpoint ckpt_130001, 30000 iterations, batch size 10, patches of 17x256x256 voxels, learning rate 0.003, validation every 2000 steps, the checkpoint of the last step (ckpt_030000) is taken.

Stage 4 (s4), released as Rectina m2: from the s3 checkpoint of the last step, 20000 iterations, batch size 10, patches of 17x384x384 voxels, learning rate 0.003. The checkpoint released is the one of the last step (ckpt_020000).

Two additions apply to stage 1 only. The first is the per patch depth warp, which replaces villa's flat random shift of the depth window by a smooth random shift of the planes read for each training patch: a smooth field built from Gaussian noise at scales of 8, 16, 32, 64 pixels, rounded to the nearest plane, with a maximum offset of 3 planes; empty rectangles of 32 pixels are blanked, with probability 0.05, in the image and in its supervision mask; labels are read without shift. The registration of the chain sets the range to 2, the maximum offset to 3 and the amplitude to 2. Stages 3 and 4 carry no depth warp.

The second is the robust_mad guard of the training hook. Villa's robust_mad normalisation divides a patch by its scaled median absolute deviation and falls back to a different scale only when that deviation is under 1e-6. A warped window that is about half fill (value 0) has a median at or near 0 and a tiny deviation, so the patch would be divided by a very small scale and reach huge values. The guard changes this for the image the warp has just built, and only in the robust_mad mode: when the scaled deviation of the patch, after clipping to the 1st and 99th percentiles, is at least that floor but under 0.01 of the span between those percentiles, the patch is normalised with villa's own fallback branch instead of by the tiny scale; in every other case villa normalises as before. The guard can act only where a warp has just built the image, that is in stage 1.

Nothing else of the recipe is dropped or added after stage 1: the depth warp is dropped from stages 3 and 4, and the checkpoint of the last step of stage 4 is the one released as Rectina m2. Settings common to the three stages: mixed precision fp16, scheduler diffusers_cosine_warmup, weight decay 3e-05, label smoothing of the BCE loss 0.5, seed 42, 32 validation batches per evaluation.

VI. Measurement

The maps of every model were produced on graphics processors in rented pods, and the scores were computed on CPU from those maps; no read of any model on CPU enters a table.

[1] Windows: the windows table of the bench (`five-model-bench/evidence`, listed in appendix C), fixed before any map existed.

[2] The eight held-out units are the validation-only units of the val54 group of the windows table of the bench, listed in the file `held-out-units.csv` of the evidence of this article.

[3] The first training step was continued in place from iteration 79001 of an earlier run of the same recipe, itself trained from scratch.

During this work comparisons were run against other published readers. This article reports the comparison of Rectina m1 with Reader v2, the earlier published reader, and with Reader v3, the published reader of the project, and that of Rectina m2 with Reader v3.

We score every reader with the area under the ROC curve of its ink probability against the organiser mask of the unit, taken on the window of the unit that was fixed from the labels before any map existed. The eight held-out units fall in two energy blocks: 6 units scanned at 116 keV and 2 units scanned at 113 keV.[4] Each of the tables in the sections that follow gives for all fourteen units, in three blocks, three measures of the two readers it compares (the AUC, the AUC on thin strokes and false ink at 80 per cent recall) and, under the units, the mean and the median of each measure. For stages 3 and 4 the checkpoint is the last step of the stage, as for Reader v3; for stage 1 it is the checkpoint at its fixed step, where the run was stopped (the stage was set for 131000 iterations and stopped at the step 130001); no checkpoint of a stage was selected on the validation set; which variant of the procedure is reported was chosen on the fourteen units (section VIII), so the paired tests of this article are exploratory and descriptive, not confirmatory.

As a check of the pods, 3 GPU maps of Reader v2, on 2 distinct units (some units were read more than once), were compared with the stored CPU maps of the same units against the limits declared before any number (largest absolute difference 0.05, largest mean absolute difference 0.005, smallest Pearson correlation 0.995); the largest absolute difference observed is 0.00156, the largest mean absolute difference 6.74e-05, the smallest Pearson correlation 0.9999998, and 3 of the 3 maps are within the limits. Likewise 3 GPU maps of Reader v3, on 2 distinct units (some units were read more than once), were compared with the stored CPU maps of the same units against the limits declared before any number (largest absolute difference 0.05, largest mean absolute difference 0.005, smallest Pearson correlation 0.995); the largest absolute difference observed is 0.00277, the largest mean absolute difference 6.97e-05, the smallest Pearson correlation 0.9999977, and 3 of the 3 maps are within the limits. No stored CPU map exists for Rectina m1 and Rectina m2, which were read only on graphics processors, so no parity claim is made for them.

VII. Results for Rectina m1 against Reader v2

Reader v2 is the reader that Rectina m1 is set against first. Table I gives the three measures of Reader v2 and Rectina m1 on the fourteen units, in the same format as the other tables.

Over the 8 held-out units Rectina m1 scores above Reader v2 on 8 of 8 and the median difference is +0.0270, above zero; over the 6 control units it scores above Reader v2 on 4 of 6 and the median difference is +0.0104, so that on the controls the median of Rectina m1 is above that of Reader v2.

On the eight held-out units the mean score is 0.8421 for Reader v2 and 0.8696 for Rectina m1; Rectina m1 scores above Reader v2 on 8 of the 8 units, the median difference is +0.0270 and the mean difference is +0.0275. On the six control units Rectina m1 scores above Reader v2 on 4 of 6 with a median difference of +0.0104. On the four controls with organisers' labels (ag1, ag2, w00, 0343), a subset of the six, the split following from who made the labels of each unit, a fact about the data and not a choice made on the scores (both counts are given because the split was not fixed before the scores), the mean score is 0.8596 for Reader v2 and 0.8694 for Rectina m1; Rectina m1 scores above Reader v2 on 3 of the 4 units, the median difference is +0.0104 and the mean difference is +0.0098, and the unit that lowers the mean is ag1 (-0.0015).

At the threshold that marks 80 per cent of the ink pixels of each unit, Rectina m1 marks less blank papyrus as ink than Reader v2 on 8 of 8 units (mean share 0.2284 against 0.2865, median 0.2436 against 0.2768). There is no unit where Rectina m1 marks more blank papyrus as ink than Reader v2 at that threshold.

Exploratory paired tests of the AUC difference, descriptive and not confirmatory (Rectina m1 minus Reader v2), the unit being the pair, zero differences dropped before the ranks (zero_method wilcox of scipy 1.18.1), exact p-values read at build time from the pinned statistics file (sha256 278491a321cd). Over the 8 held-out units (n = 8 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.00391 one-sided (Rectina m1 greater than Reader v2) and 0.00781 two-sided, and the sign test on the 8 of 8 non zero differences where Rectina m1 is above gives 0.00391 one-sided and 0.00781 two-sided. Over the 6 control units (n = 6 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.109 one-sided (Rectina m1 greater than Reader v2) and 0.219 two-sided, and the sign test on the 4 of 6 non zero differences where Rectina m1 is above gives 0.344 one-sided and 0.688 two-sided. Over all 14 units (n = 14 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.00061 one-sided (Rectina m1 greater than Reader v2) and 0.00122 two-sided, and the sign test on the 12 of 14 non zero differences where Rectina m1 is above gives 0.00647 one-sided and 0.0129 two-sided. Rectina m1 was trained with a single seed (42), so the spread between two runs of the same recipe is not measured here, and the paired test above concerns this checkpoint, not the recipe.

VIII. Results for Rectina m2 against Reader v3

Table II gives for all fourteen units, in three blocks, three measures of Reader v3 and Rectina m2 (the AUC, the AUC on thin strokes and false ink at 80 per cent recall) and, under the units, the mean and the median of each measure; the text below it gives, for the eight held-out units, the mean, the median and the number of units on which Rectina m2 scores above Reader v3 in AUC.

The rule for adopting the model was fixed in advance: on the 8 held-out units the count of units on which the model scores above Reader v3 must be at least 6 of 8, and the median gain over Reader v3 must be above 0.003. The rule is evaluated on the held-out units only. On the 8 held-out units the count

[4]Energy blocks: the file `held-out-units.csv` of the evidence of this article.

TABLE I

Three measures of Rectina m1 against Reader v2 on the fourteen units, target a: the AUC, the AUC on thin strokes and false ink at 80 per cent recall. The cell of Rectina m1 gives in parentheses the signed difference against Reader v2. The rows Mean and Median are taken over all the units of the table that are not left out; the block of the community labels is information only.

unit	AUC		AUC on thin strokes		false ink at 80 per cent recall	
	Reader v2	Rectina m1	Reader v2	Rectina m1	Reader v2	Rectina m1
Control units outside the training segments, organisers' labels: 0343 (approved letters, 116 keV); 0841 (113 keV)						
0343	0.8956	▲ **0.9105** (+0.0149)	0.9136	▲ **0.9246** (+0.0110)	0.0807	▲ **0.0705** (-0.0103)
0841 ag1	**0.8410**	▼ 0.8395 (-0.0015)	**0.8538**	▼ 0.8519 (-0.0020)	**0.2379**	▼ 0.2530 (+0.0151)
0841 ag2	0.8250	▲ **0.8449** (+0.0199)	0.8431	▲ **0.8600** (+0.0169)	0.2904	▲ **0.2631** (-0.0273)
0841 w00	0.8767	▲ **0.8826** (+0.0059)	0.8964	▲ **0.8997** (+0.0033)	**0.1912**	▼ 0.1939 (+0.0027)
Control unit outside the training segments, community labels, information only: 1447 (116 keV)						
1447 w058	**0.8575**	▼ 0.8491 (-0.0083)	**0.8934**	▼ 0.8865 (-0.0069)	**0.2689**	▼ 0.2813 (+0.0123)
1447 w060	0.7777	▲ **0.7958** (+0.0181)	0.8185	▲ **0.8429** (+0.0243)	0.4240	▲ **0.3770** (-0.0470)
Held-out segments of training scrolls, our validation labels: 0009B and 0343P (116 keV), 0500P2 (113 keV)						
0009B·172639	0.8773	▲ **0.9259** (+0.0486)	0.8849	▲ **0.9303** (+0.0454)	0.2132	▲ **0.0960** (-0.1172)
0009B·125754	0.9359	▲ **0.9600** (+0.0241)	0.9470	▲ **0.9663** (+0.0192)	0.0802	▲ **0.0455** (-0.0347)
0009B·133208	0.8473	▲ **0.8773** (+0.0300)	0.8595	▲ **0.8873** (+0.0278)	0.2689	▲ **0.2203** (-0.0486)
0009B·140325	0.8767	▲ **0.8784** (+0.0016)	0.8858	▲ **0.8860** (+0.0002)	0.2182	▲ **0.1908** (-0.0274)
0500P2·055229	0.8396	▲ **0.8605** (+0.0209)	0.8527	▲ **0.8698** (+0.0171)	0.3411	▲ **0.2854** (-0.0556)
0500P2·055230	0.7602	▲ **0.7984** (+0.0382)	0.7711	▲ **0.8078** (+0.0367)	0.4502	▲ **0.3541** (-0.0961)
0343P·233748-0	0.8383	▲ **0.8584** (+0.0201)	0.8433	▲ **0.8629** (+0.0196)	0.2847	▲ **0.2669** (-0.0178)
0343P·172054-1	0.7615	▲ **0.7981** (+0.0365)	0.7696	▲ **0.8035** (+0.0339)	0.4359	▲ **0.3679** (-0.0680)
Mean	0.8436	▲ **0.8628** (+0.0192)	0.8595	▲ **0.8771** (+0.0176)	0.2704	▲ **0.2333** (-0.0371)
Median	0.8441	▲ **0.8595** (+0.0153)	0.8567	▲ **0.8779** (+0.0212)	0.2689	▲ **0.2580** (-0.0109)

AUC: the rank AUC of ink against blank pixels.
AUC on thin strokes: the rank AUC of the central 80 per cent of each stroke against blank pixels at least 207 um from any label, the band between not scored.
false ink at 80 per cent recall: the share of blank pixels marked as ink at the threshold that recalls 80 per cent of the ink pixels, where less is better.
For the controls the AUC is taken against the far negatives of the bench; for the held-out units against all blank pixels of the validation mask.
The better value of each pair in bold; a green triangle pointing up before Rectina m1 where it is better, a red triangle pointing down where it is worse.
A pair whose two values are equal at four decimals has no bold value and no triangle.

above Reader v3 is 6 of 8 and the median gain is +0.0137, so the stated rule is met. Over all 14 units the count above Reader v3 is 11 of 14. Of the six variants scored on the fourteen units, four meet this rule against Reader v3 and six against Reader v2, so the rule is descriptive, like the rest.

Over the eight units the mean score is 0.8547 for Reader v3 and 0.8704 for Rectina m2, and the medians are 0.8503 and 0.8754. Rectina m2 scores above Reader v3 on 6 of the 8 units, the median difference is +0.0137 and the mean difference is +0.0157, and the unit that lowers the mean is 20250919140325 (-0.0082). In the 116 keV block the count is 5 of 6, and in the 113 keV block it is 1 of 2. The count of held-out units above Reader v3 is 6 of 8, and the median over the eight units of the score of Rectina m2 minus the score of Reader v3 is +0.0137.

Over the 14 units, 8 held-out and 6 control, two of which are information units, Rectina m2 is above Reader v3 on 11 and the mean difference is +0.0104.

At the threshold that marks 80 per cent of the ink pixels of each unit, Rectina m2 marks less blank papyrus as ink than Reader v3 on 5 of 8 units (mean share 0.2267 against 0.2585, median 0.2306 against 0.2917). The units where Rectina m2 marks more blank papyrus as ink than Reader v3 at that threshold are 20250716055229, 20250904233748-0_b2 and 20250919140325; the units where its area under the curve is below that of Reader v3 are 20250716055229 and 20250919140325, which is not the same set (only in the first: 20250904233748-0_b2; only in the second: no unit).

Exploratory paired tests of the AUC difference, descriptive and not confirmatory (Rectina m2 minus Reader v3), the unit being the pair, zero differences dropped before the ranks (zero_method wilcox of scipy 1.18.1), exact p-values read at build time from the pinned statistics file (sha256 ae8ffe90e17a). Over the 8 held-out units (n = 8 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.0391 one-sided (Rectina m2 greater than Reader v3) and 0.0781 two-sided, and the sign test on the 6 of 8 non zero differences where Rectina m2 is above gives 0.145 one-sided and 0.289 two-sided. Over the 6 control units (n = 6 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.0781 one-sided (Rectina m2 greater than Reader v3) and 0.156 two-sided, and the sign test on the 5 of 6 non zero differences where Rectina m2 is above gives 0.109 one-sided and 0.219 two-sided. Over all 14 units (n = 14 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.0123 one-sided (Rectina m2 greater than Reader v3) and 0.0245 two-sided, and the sign test on the 11 of 14 non zero differences where Rectina m2 is above gives 0.0287 one-sided and 0.0574 two-sided. Other variants of the same procedure were also trained, and the choice was made on the same units; the paired tests describe this checkpoint, not the recipe. Besides the two checkpoints reported here, four other variants of the procedure were scored on the same fourteen units (the pinned file of scores holds them); the checkpoints reported were chosen on those units, so the paired tests describe these checkpoints

TABLE II

Three measures of Rectina m2 against Reader v3 on the fourteen units, target a: the AUC, the AUC on thin strokes and false ink at 80 per cent recall. The cell of Rectina m2 gives in parentheses the signed difference against Reader v3. The rows Mean and Median are taken over all the units of the table that are not left out; the block of the community labels is information only.

unit	AUC		AUC on thin strokes		false ink at 80 per cent recall	
	Reader v3	Rectina m2	Reader v3	Rectina m2	Reader v3	Rectina m2
Control units outside the training segments, organisers' labels: 0343 (approved letters, 116 keV); 0841 (113 keV)						
0343	0.9078	▲ **0.9136** (+0.0058)	0.9246	▲ **0.9280** (+0.0033)	0.0690	▲ **0.0671** (-0.0019)
0841 ag1	**0.8513**	▼ 0.8466 (-0.0047)	**0.8648**	▼ 0.8594 (-0.0053)	**0.2153**	▼ 0.2440 (+0.0287)
0841 ag2	0.8297	▲ **0.8364** (+0.0068)	0.8474	▲ **0.8504** (+0.0030)	0.2681	▲ **0.2413** (-0.0268)
0841 w00	0.9055	▲ **0.9146** (+0.0091)	0.9221	▲ **0.9299** (+0.0078)	0.1444	▲ **0.1360** (-0.0084)
Control unit outside the training segments, community labels, information only: 1447 (116 keV)						
1447 w058	0.8610	▲ **0.8627** (+0.0017)	0.8965	▲ **0.9014** (+0.0048)	0.2665	▲ **0.2412** (-0.0253)
1447 w060	0.7842	▲ **0.7863** (+0.0021)	0.8276	▲ **0.8340** (+0.0064)	0.4095	▲ **0.4048** (-0.0047)
Held-out segments of training scrolls, our validation labels: 0009B and 0343P (116 keV), 0500P2 (113 keV)						
0009B·172639	0.8456	▲ **0.8891** (+0.0435)	0.8515	▲ **0.8915** (+0.0400)	0.3114	▲ **0.1943** (-0.1172)
0009B·125754	0.9562	▲ **0.9734** (+0.0172)	0.9636	▲ **0.9777** (+0.0142)	0.0460	▲ **0.0235** (-0.0225)
0009B·133208	0.8551	▲ **0.8618** (+0.0066)	0.8672	▲ **0.8718** (+0.0046)	0.2774	▲ **0.2670** (-0.0104)
0009B·140325	**0.9040**	▼ 0.8958 (-0.0082)	**0.9110**	▼ 0.9028 (-0.0082)	**0.1574**	▼ 0.1715 (+0.0141)
0500P2·055229	**0.8971**	▼ 0.8903 (-0.0067)	**0.9074**	▼ 0.8978 (-0.0095)	**0.1360**	▼ 0.1573 (+0.0214)
0500P2·055230	0.7881	▲ **0.8143** (+0.0262)	0.7980	▲ **0.8222** (+0.0242)	0.3998	▲ **0.3252** (-0.0747)
0343P·233748-0	0.8340	▲ **0.8441** (+0.0101)	0.8398	▲ **0.8490** (+0.0092)	**0.3059**	▼ 0.3202 (+0.0143)
0343P·172054-1	0.7573	▲ **0.7942** (+0.0368)	0.7669	▲ **0.7987** (+0.0318)	0.4336	▲ **0.3547** (-0.0789)
Mean	0.8555	▲ **0.8659** (+0.0104)	0.8706	▲ **0.8796** (+0.0090)	0.2458	▲ **0.2249** (-0.0209)
Median	0.8532	▲ **0.8622** (+0.0090)	0.8660	▲ **0.8816** (+0.0156)	0.2673	▲ **0.2412** (-0.0261)

AUC: the rank AUC of ink against blank pixels.
AUC on thin strokes: the rank AUC of the central 80 per cent of each stroke against blank pixels at least 207 um from any label, the band between not scored.
false ink at 80 per cent recall: the share of blank pixels marked as ink at the threshold that recalls 80 per cent of the ink pixels, where less is better.
For the controls the AUC is taken against the far negatives of the bench; for the held-out units against all blank pixels of the validation mask.
The better value of each pair in bold; a green triangle pointing up before Rectina m2 where it is better, a red triangle pointing down where it is worse.
A pair whose two values are equal at four decimals has no bold value and no triangle.

and are exploratory, not confirmatory. Both Rectina m2 and Reader v3 were trained with a single seed (42 in both training configurations), so the spread between two runs of the same recipe is not measured here. The comparison is therefore between two single runs, and the paired test above concerns these two checkpoints, not the two recipes.[5]

A. The controls

The control units were scored on the whole box of the unit, against the raw organiser labels, with the same scorer.[6] None of the segments of the six control units was used in training; the overlap check compares the keys of the segments, not neighbouring areas of one volume. Four of them carry the labels of the organisers: ag1, ag2 and w00, drawn by hand by the organisers, and 0343, whose labels are the mask of the approved letters, which we know through the image of the announcement. The other two, w058 and w060, carry labels made by members of the community; the header of their block in Table II says that they are information only, which means that their labels are not those of the organisers and that they stay out of the sentence below, while they are counted in the Mean and Median rows of the table. The eight held-out units are different in kind: they are segments of scrolls that were used in training, and the header of their block calls them held-out segments of training scrolls with our validation labels.

On the six control units Rectina m2 scores above Reader v3 on 5 of 6 with a median difference of +0.0040. On the four controls with organisers' labels (ag1, ag2, w00, 0343), a subset of the six, the split following from who made the labels of each unit, a fact about the data and not a choice made on the scores (both counts are given because the split was not fixed before the scores), the mean score is 0.8736 for Reader v3 and 0.8778 for Rectina m2, and the medians are 0.8784 and 0.8801. Rectina m2 scores above Reader v3 on 3 of the 4 units, the median difference is +0.0063 and the mean difference is +0.0042, and the unit that lowers the mean is ag1 (-0.0047).

The six control figures show one control unit each, in three rows of two panels.

IX. Results for Rectina m1 against Reader v3

Rectina m1 is the checkpoint at the end of the first stage. Table III gives, in the format of the table of the previous section, the three measures of Reader v3 and Rectina m1 on the fourteen units.

Over the 8 held-out units Rectina m1 scores above Reader v3 on 6 of 8 and the median difference is +0.0162, above zero; over the 6 control units it scores above Reader v3 on 3 of 6

[5]The seed of Reader v3 is the key `seed` of `reader-v3/train_config.json`, value 42; the seed of Rectina m2 is the key `seed` of the job entry of its training chain in the job record of that run, value 42.

[6]Control windows: the windows table of the bench, rows of kind control, scorer thmb7 on the raw organiser labels, whole box.

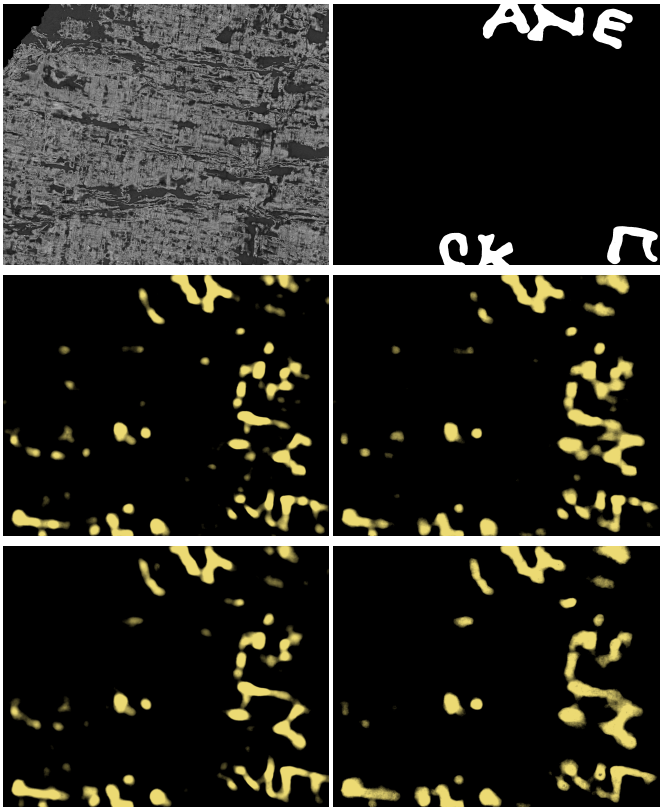

Fig. 1. Control unit 0841 ag1, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.447222 to 0.621051 (ramp); Rectina m1 0.623262 to 0.741702 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.470294 to 0.693186 (ramp); Rectina m2 0.671638 to 0.774498 (ramp)). Each map is stretched between its own percentiles. Reader and model stand side by side in the rows 2 and 3.

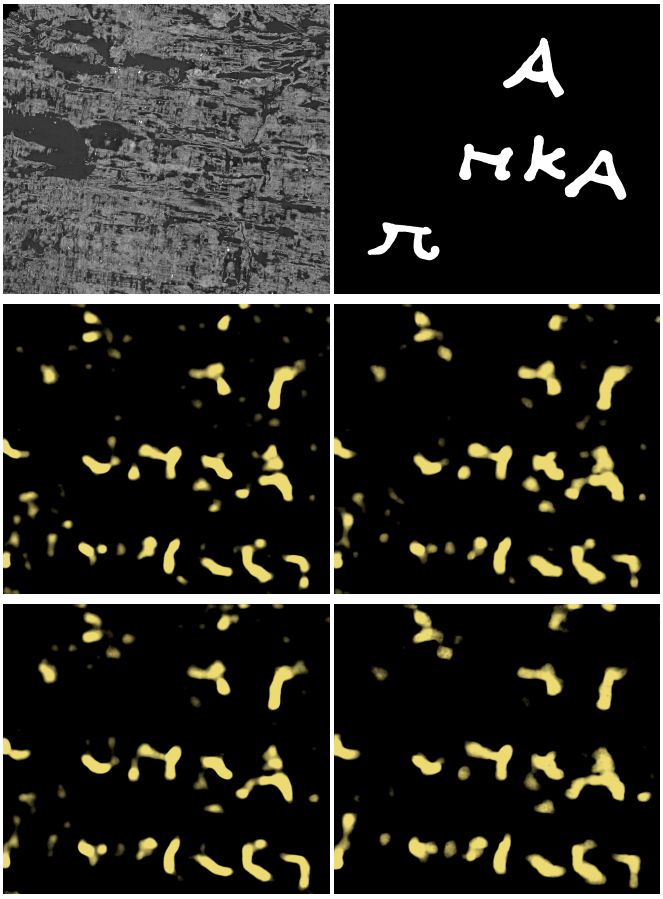

Fig. 2. Control unit 0841 ag2, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.404870 to 0.589240 (ramp); Rectina m1 0.560881 to 0.723274 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.431564 to 0.660926 (ramp); Rectina m2 0.615411 to 0.768217 (ramp)). Each map is stretched between its own percentiles. Reader and model stand side by side in the rows 2 and 3.

and the median difference is -0.0046, so that on the controls the median of Rectina m1 is below that of Reader v3.

On the eight held-out units the mean score is 0.8547 for Reader v3 and 0.8696 for Rectina m1; Rectina m1 scores above Reader v3 on 6 of the 8 units, the median difference is +0.0162 and the mean difference is +0.0150, and the unit that lowers the mean is 20250716055229 (-0.0366). On the six control units Rectina m1 scores above Reader v3 on 3 of 6 with a median difference of -0.0046. On the four controls with organisers' labels (ag1, ag2, w00, 0343), a subset of the six, the split following from who made the labels of each unit, a fact about the data and not a choice made on the scores (both counts are given because the split was not fixed before the scores), the mean score is 0.8736 for Reader v3 and 0.8694 for Rectina m1; Rectina m1 scores above Reader v3 on 2 of the 4 units, the median difference is -0.0046 and the mean difference is -0.0042, and the unit that lowers the mean is w00 (-0.0229).

At the threshold that marks 80 per cent of the ink pixels of each unit, Rectina m1 marks less blank papyrus as ink than Reader v3 on 6 of 8 units (mean share 0.2284 against 0.2585, median 0.2436 against 0.2917). The units where Rectina m1 marks more blank papyrus as ink than Reader v3 at that threshold are 20250716055229 and 20250919140325, and these are also the units where its area under the curve is below that of Reader v3.

Exploratory paired tests of the AUC difference, descriptive and not confirmatory (Rectina m1 minus Reader v3), the unit being the pair, zero differences dropped before the ranks (zero_method wilcox of scipy 1.18.1), exact p-values read at build time from the pinned statistics file (sha256 ae8ffe90e17a). Over the 8 held-out units (n = 8 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.191 one-sided (Rectina m1 greater than Reader v3) and 0.383 two-sided, and the sign test on the 6 of 8 non zero differences where Rectina m1 is above gives 0.145 one-sided and 0.289 two-sided. Over the 6 control units (n = 6 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.719 one-sided (Rectina m1 greater than Reader v3) and 0.688 two-sided, and the sign test on the 3 of 6 non zero differences where Rectina m1 is above gives 0.656 one-sided and 1 two-sided. Over all 14 units (n = 14 after dropping 0 zero differences) the exact Wilcoxon signed-rank p is 0.292 one-sided (Rectina m1 greater than Reader v3) and 0.583 two-sided, and the sign test on the 9 of 14 non zero differences where Rectina m1 is above gives 0.212 one-sided and 0.424

TABLE III
Three measures of Rectina m1 against Reader v3 on the fourteen units, target a: the AUC, the AUC on thin strokes and false ink at 80 per cent recall. The cell of Rectina m1 gives in parentheses the signed difference against Reader v3. The rows Mean and Median are taken over all the units of the table that are not left out; the block of the community labels is information only.

unit	AUC		AUC on thin strokes		false ink at 80 per cent recall	
	Reader v3	Rectina m1	Reader v3	Rectina m1	Reader v3	Rectina m1
Control units outside the training segments, organisers' labels: 0343 (approved letters, 116 keV); 0841 (113 keV)						
0343	0.9078	▲ **0.9105** (+0.0026)	0.9246	0.9246 (+0.0000)	**0.0690**	▼ 0.0705 (+0.0014)
0841 ag1	**0.8513**	▼ 0.8395 (-0.0119)	**0.8648**	▼ 0.8519 (-0.0129)	**0.2153**	▼ 0.2530 (+0.0376)
0841 ag2	0.8297	▲ **0.8449** (+0.0153)	0.8474	▲ **0.8600** (+0.0126)	0.2681	▲ **0.2631** (-0.0050)
0841 w00	**0.9055**	▼ 0.8826 (-0.0229)	**0.9221**	▼ 0.8997 (-0.0223)	**0.1444**	▼ 0.1939 (+0.0495)
Control unit outside the training segments, community labels, information only: 1447 (116 keV)						
1447 w058	**0.8610**	▼ 0.8491 (-0.0118)	**0.8965**	▼ 0.8865 (-0.0100)	**0.2665**	▼ 0.2813 (+0.0148)
1447 w060	0.7842	▲ **0.7958** (+0.0116)	0.8276	▲ **0.8429** (+0.0153)	0.4095	▲ **0.3770** (-0.0325)
Held-out segments of training scrolls, our validation labels: 0009B and 0343P (116 keV), 0500P2 (113 keV)						
0009B·172639	0.8456	▲ **0.9259** (+0.0803)	0.8515	▲ **0.9303** (+0.0788)	0.3114	▲ **0.0960** (-0.2154)
0009B·125754	0.9562	▲ **0.9600** (+0.0038)	0.9636	▲ **0.9663** (+0.0027)	0.0460	▲ **0.0455** (-0.0006)
0009B·133208	0.8551	▲ **0.8773** (+0.0222)	0.8672	▲ **0.8873** (+0.0201)	0.2774	▲ **0.2203** (-0.0571)
0009B·140325	**0.9040**	▼ 0.8784 (-0.0256)	**0.9110**	▼ 0.8860 (-0.0250)	**0.1574**	▼ 0.1908 (+0.0334)
0500P2·055229	**0.8971**	▼ 0.8605 (-0.0366)	**0.9074**	▼ 0.8698 (-0.0376)	**0.1360**	▼ 0.2854 (+0.1494)
0500P2·055230	0.7881	▲ **0.7984** (+0.0103)	0.7980	▲ **0.8078** (+0.0098)	0.3998	▲ **0.3541** (-0.0458)
0343P·233748-0	0.8340	▲ **0.8584** (+0.0244)	0.8398	▲ **0.8629** (+0.0231)	0.3059	▲ **0.2669** (-0.0390)
0343P·172054-1	0.7573	▲ **0.7981** (+0.0407)	0.7669	▲ **0.8035** (+0.0366)	0.4336	▲ **0.3679** (-0.0657)
Mean	0.8555	▲ **0.8628** (+0.0073)	0.8706	▲ **0.8771** (+0.0065)	0.2458	▲ **0.2333** (-0.0125)
Median	0.8532	▲ **0.8595** (+0.0062)	0.8660	▲ **0.8779** (+0.0119)	0.2673	▲ **0.2580** (-0.0093)

AUC: the rank AUC of ink against blank pixels.
AUC on thin strokes: the rank AUC of the central 80 per cent of each stroke against blank pixels at least 207 um from any label, the band between not scored.
false ink at 80 per cent recall: the share of blank pixels marked as ink at the threshold that recalls 80 per cent of the ink pixels, where less is better.
For the controls the AUC is taken against the far negatives of the bench; for the held-out units against all blank pixels of the validation mask.
The better value of each pair in bold; a green triangle pointing up before Rectina m1 where it is better, a red triangle pointing down where it is worse.
A pair whose two values are equal at four decimals has no bold value and no triangle.

two-sided. Both Rectina m1 and Reader v3 were trained with a single seed (42 in both training configurations), so the spread between two runs of the same recipe is not measured here. The comparison is therefore between two single runs, and the paired test above concerns these two checkpoints, not the two recipes.[7]

X. Usage

The pieces for using the checkpoints are published with this article in its results folder [2]: the reader in `results/db27c806-rectina/src/read`, the recipe of the render in `results/db27c806-rectina/src/render` and the converters of a map in `results/db27c806-rectina/src/convert`. This section says what each piece does and what was run and not run; the README of each folder says more.

Input. A checkpoint of Rectina reads a surface volume: a stack of layers sampled along the normal of a segmented surface, stored as an array of shape (planes, rows, columns), as an npy file or an OME-Zarr folder. The reader takes seventeen consecutive planes from the stack and slides a square patch over rows and columns. The stacks used in this work have twenty three planes, stored as uint16 holding the rendered grey value times 256, so that 65280 is the saturated value of an 8 bit render, and divided by 65280 before reading; the seven read starts are the seven windows of seventeen planes inside the twenty three. The reader applies to every patch the robust median absolute deviation normalisation of the villa ink detection code (percentiles 1 and 99). The stack has to be rendered along the surface normal at the scale of the training data, about 9 um. Reads of stacks from another render were not tested.

Reading a unit. The reader of `results/db27c806-rectina/src/read` is run with the villa package at the commit that its README pins (the commands of the README):

```
git clone https://github.com/ScrollPrize/villa.git &&
↪ git -C villa checkout
↪ 0bc2c241de7e289e36459468a9a0b32257679966
python3 rectina_read.py info    --checkpoint
↪ m2/ckpt_020000.pth
python3 rectina_read.py read    --stack unit.npy
↪ --checkpoint m2/ckpt_020000.pth --villa-src
↪ villa/vesuvius/src --device cuda --out out/unit
python3 rectina_read.py compare out/unit/map.npy
↪ reference_map.npy
```

Each start reads its seventeen planes with two views (the identity and the view turned by 180 degrees, whose maps are averaged), four lattices shifted by a quarter of a patch, bfloat16 autocast on a graphics processor and Hann window blending; the map is the mean over the seven starts. The patch is 128 pixels for Rectina m1 and 384 pixels for Rectina m2: it is the crop size of the configuration embedded in each checkpoint, read from the checkpoint (not inferred). A read needs a graphics processor in practice: for Rectina m2 a read of

[7]The seed of Reader v3 is the key `seed` of `reader-v3/train_config.json`, value 42; the seed of Rectina m1 is the key `seed` of the job entry of its training chain in the job record of that run, value 42.

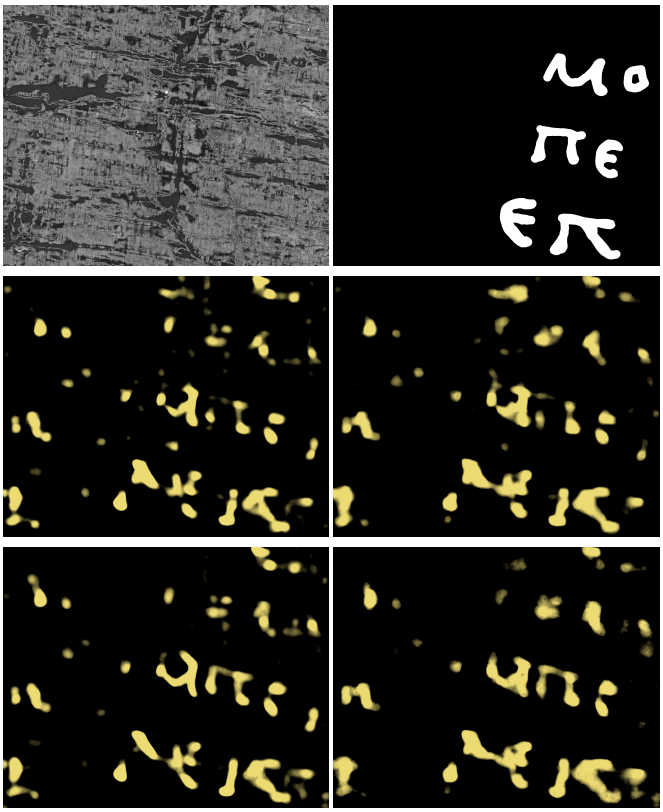

Fig. 3. Control unit 0841 w00, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.419276 to 0.595756 (ramp); Rectina m1 0.577431 to 0.724182 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.454060 to 0.672485 (ramp); Rectina m2 0.650603 to 0.768363 (ramp)). Each map is stretched between its own percentiles. Reader and model stand side by side in the rows 2 and 3.

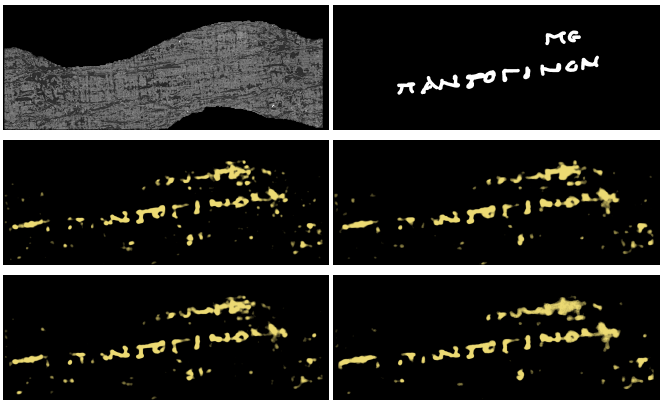

Fig. 4. Control unit 0343, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.409836 to 0.590749 (ramp); Rectina m1 0.548366 to 0.721975 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.444697 to 0.678368 (ramp); Rectina m2 0.627931 to 0.765628 (ramp)). Each map is stretched between its own percentiles. The control is shown rotated 180 degrees, as the index of the panels records. Reader and model stand side by side in the rows 2 and 3.

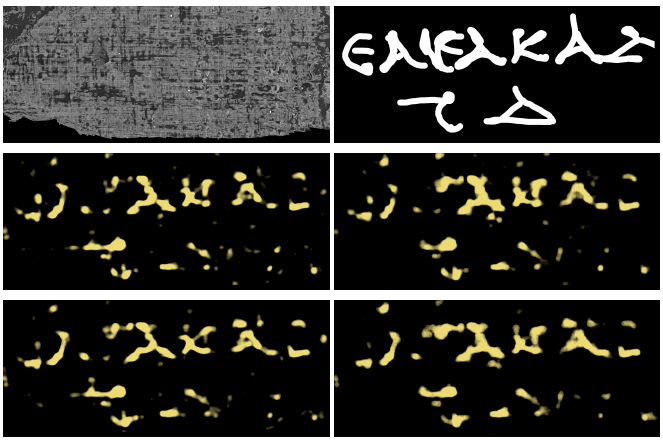

Fig. 5. Control unit 1447 w058, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.385394 to 0.561662 (ramp); Rectina m1 0.533701 to 0.711716 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.450583 to 0.670398 (ramp); Rectina m2 0.625307 to 0.773366 (ramp)). Each map is stretched between its own percentiles. Reader and model stand side by side in the rows 2 and 3.

one million pixels took about 102 seconds for the seven starts on a graphics processor of type 4090 (peak memory 7631 MiB), and for Rectina m1 about 80 seconds (peak memory 967 MiB), both read from the records of the reads of unit ag1. Run by: the author, on a rented RTX 4090 machine with bfloat16, for unit ag1 only, the commands read and compare of the README; the result is that the public reader reproduces the stored maps of the same checkpoints on this one unit (the sha256 of both maps equal the stored ones), and nothing more: it says nothing about other units, other stacks or other hardware. Also run: the comparison of a CPU read of a crop of unit ag1 with Rectina m1 against the stored map, inside the declared limits. Not tested: the command info, Rectina m2 on a CPU, other graphics processors.

Output. For each unit the read writes map.npy (float32, rows by columns of the stack, values between 0 and 1), reads.csv and a record of the sha256 of the outputs. The map is a ranking score of ink against papyrus and not a calibrated probability: the reader of Reader v3 says of its own values that “they are a relative ink score, not a calibrated probability” (README of its repository, line 144), and the same holds for the maps of Rectina, which come from a model trained with label smoothing. The recipe uses label smoothing of 0.5 on the binary cross entropy, which sets the training targets to 0.25 for background and 0.75 for ink; the ink detection tutorial states it (“the training targets are 0.25 for background and 0.75 for ink” [8]). This is deliberate, and it is why the values of a map occupy a compressed band. On a real unit the values span a narrow band, for instance 0.23 to 0.84 for Rectina m2 on ag1 and 0.00 to 0.85 on 0343, so a fixed threshold such as 0.5 means nothing; evaluate by rank (the area under the curve) or choose a threshold on a labelled region. To make an image, scale the map linearly between two percentiles and write an 8 bit tif.

Worked example. Unit ag1, a control unit of the comparison (organisers’ human labels, none of its segments used in

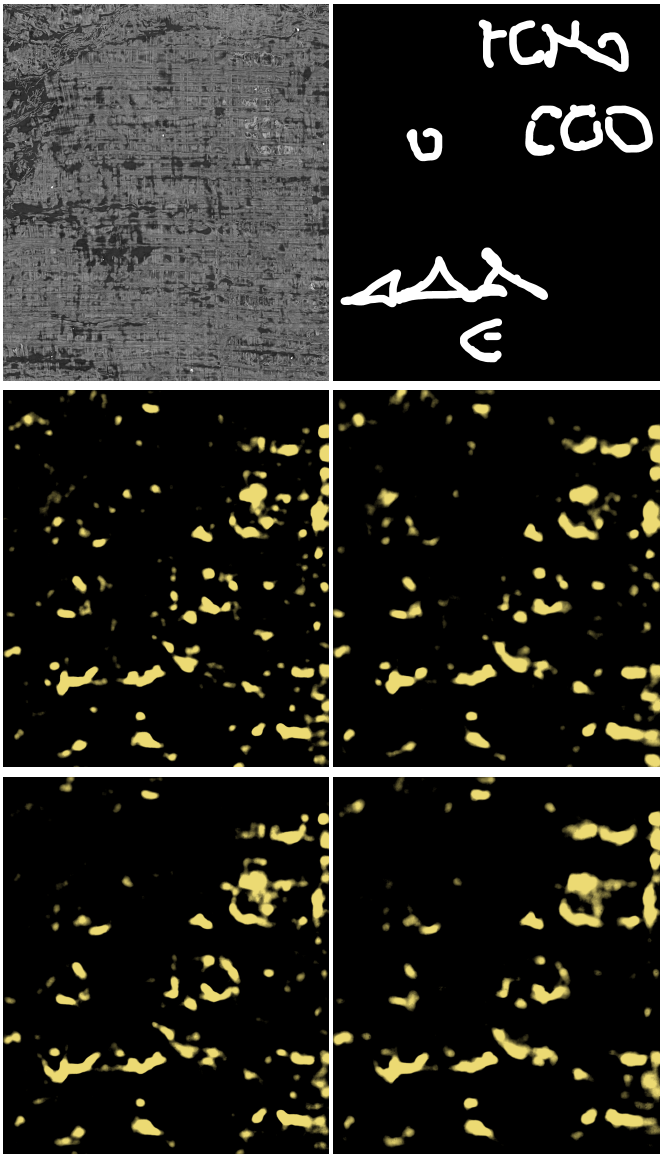

Fig. 6. Control unit 1447 w060, three rows of two panels. Row 1: left, Texture; right, Labels. Row 2: left, Reader v2; right, Rectina m1 (display windows: Reader v2 0.358837 to 0.460831 (ramp); Rectina m1 0.419819 to 0.580751 (ramp)). Row 3: left, Reader v3; right, Rectina m2 (display windows: Reader v3 0.381681 to 0.538574 (ramp); Rectina m2 0.465540 to 0.657352 (ramp)). Each map is stretched between its own percentiles. Reader and model stand side by side in the rows 2 and 3.

training), 2048 x 2560 pixels. The stored map of Rectina m1 has sha256 `fc4d29c34cebd2694d53297271ad329cf3072d4588329c6ae6bd29ec827edfb5` and scores an area under the curve of 0.8395 (0.8394838188166195) against the labels with the scorer of this work; the stored map of Rectina m2 has sha256 `55e2252f660fcd4ed8551fd97cebb7d6ed56847eaebe6c7ca652ebd4e1783e55` and scores 0.8466 (0.8466005527037327); both numbers are the cells of the comparison tables. Scoring a stored map needs no graphics processor. The declared limits for a map read on another device against a reference map are a largest absolute difference of 0.05, a mean absolute difference of 0.005 and a Pearson correlation of at least 0.995.

Converting a map. The converter of `results/db27c806-rectina/src/convert` writes a map as a tif or as OME-Zarr and reads either back to check it against the npy (the commands of its README):

```
python3 rectina_map_convert.py tif map.npy
↪  map_float32.tif
python3 rectina_map_convert.py tif map.npy
↪  map_uint16.tif --dtype uint16
python3 rectina_map_convert.py zarr map.npy map_v3.zarr
python3 rectina_map_convert.py zarr map.npy map_v2.zarr
↪  --zarr-format 2 --pixel-size-um 9.36
python3 rectina_map_convert.py check map.npy
↪  map_float32.tif
python3 rectina_map_convert.py check map.npy
↪  map_uint16.tif
python3 rectina_map_convert.py check map.npy
↪  map_v3.zarr
python3 rectina_map_convert.py check map.npy
↪  map_v2.zarr
```

Run by: the test of that folder, on three real maps (two control units and one held-out unit): all four conversions and all four round trip checks passed. Not run: any OME-Zarr validator.

Rendering. The recipe of the render is in `results/db27c806-rectina/src/render`: the villa tool vc_render_tifxyz built from an upstream commit plus one patch of the study, with the settings of the stacks (twenty three planes at step one, float32 samples stored as uint16). It is a record of a recipe and does not replace the render component of the public chain. The binary of the original build is no longer on disk (a cleanup removed it); a rebuild from the same commit, patch and recipe has another file hash, and reproduced a stored render shard bit for bit in one reference run; the stacks of this recipe were not compared for equality with those of the public render component. Not tested: any other render of the same surface as input to the checkpoints.

Plugging in. The architecture is `vesuvius_unet_3d_stem_2d` with a seventeen plane crop and 508 tensors. The ink detection package of villa builds it from the configuration embedded in the checkpoint (the function that makes the model, models/model.py of the package, which names this architecture). Verified by reading code only. Not tested: the inference command of that package with the checkpoints of this article, the current villa head.

What was not tested. Other renders, other scrolls than the fourteen units of the comparison, other graphics processors than the ones of the registered runs, meshes or surfaces that are not tifxyz, and scales other than the 9 um class of the training data.

XI. Upstream contributions

The simpler rule was proposed to villa as pull request 2396, opened as a draft on 31 October 2026, not part of these models. The pull requests below are those that the models used. Each one links to its page, its state was read from the code host at the time this text was built (fetched 2026-11-01T02:06:02Z, read from the code host API, raw answer sha256 48e0cd2cb963), and the last two columns say what the change does and what role it has for the models.

The label regrid offset we found and reported in villa issue 2243 was corrected in the labels used here; the code that carried it was later removed from villa in pull request 2358, so the correction needs no pull request. The month's other contributions are listed in a separate submission.

XII. Limits

The score is our own metric, the area under the curve of a probability map against a mask, and it is not the metric of any contest. On the held-out units it is computed on one window per unit, and the choice of the window changes it; on the control units it is computed on the whole box. The labels of the controls are of three kinds that should not be mixed in a reading: human labels of the organisers for ag1, ag2 and w00, the mask of the approved letters for 0343, which we know only through the announcement image and therefore only as that image shows it, and labels of the community for w058 and w060. None of them is a reference of the same standing as the others. The 113 keV block holds two units of a single scroll scanned at another energy than the other six, so its counts are small and support no statement about energy. Each checkpoint was trained once, and no repeat of the training exists, so that the spread between two runs of the same configuration is not measured here and a difference of a few thousandths between two scores must not be read as a result. The overlap check of appendix C finds all fourteen units outside the training sets; it compares the keys of the segments, not neighbouring areas of one volume. The sources of the label set and the rule that turned them into masks are not recorded in a pinned file of this work; what is recorded is the inventory of its 103 units by kind, in the inventory of the study.

XIII. Release

The checkpoint released under this name is the last step of each stage, as for Reader v3. The models are published in the repository evilaliv3/rectina on Hugging Face, Rectina m1 in the folder m1 and Rectina m2 in the folder m2. The weights and the model cards are released under CC BY-NC 4.0 for non-commercial research use with attribution, and the code of the recipe under MIT. The panels, the sources and the numbers files of this article are published in its own results folder [2], whose `results/db27c806-rectina/src/read`, `results/db27c806-rectina/src/render` and `results/db27c806-rectina/src/convert` hold the public reader, the recipe of the render and the converters of a map (section X); the checkpoints are in the model repository [1]. Data terms: the data of the Vesuvius Challenge are subject to its data licence (dl.ash2txt.org LICENSE.txt). The figure panels show the CT surface, the human ink labels of the evaluation units and the model maps, with attribution to their authors; they are published on the author's own responsibility as figures of this work. This article and its results folder hold no data volumes and no label files.

Appendix A
The recipe table

TABLE IV
The pull requests of the project that these models used.

item	state	what it does	role in the models
villa 2396	open, draft	Falls back to villa's own normalisation when a robust_mad window is about half fill, instead of dividing by a near zero scale.	The same idea as the MAD guard of step 1, in a simpler form for villa's own normalisation; the model was trained with our guard.
villa 2247	open	Adds an opt in bilinear blend of node normals for linear surface interpolation.	The change equivalent to the blended normals of our render patch, which produced every render the model was trained and scored on.
villa issue 2243	open	grid-node offset when surfaces are moved between voxel sizes (about 7.4 px)	found and reported by us, corrected in our labels; the code path was removed in villa pull request 2358 (commits 9ddd16d23 and c73ccbbc9)

step	setting	value	source
1 s1	stage base	villa_hybrid	training job record
1 s1	iterations (to the fixed step)	131000	training job record
1 s1	batch (patches per step, sum of the quotas)	64	training job record
1 s1	per-scroll quotas of the batch	{"0139": 12, "1667": 8, "Paris4": 6, "0814": 7, "0009B": 8, "0343P": 5, "0500P2": 7, "0172": 11}	training job record
1 s1	mixed precision	fp16	stage configuration
1 s1	patch size (z, y, x)	[17, 128, 128]	stage configuration
1 s1	learning rate	0.01	stage configuration
1 s1	optimizer	sgd	stage configuration
1 s1	warmup steps	1000	stage configuration
1 s1	validation every	1000	training job record
1 s1	depth warp augmentation max offset	3	training job record
1 s1	checkpoint taken at the end of the step	ckpt_130001.pth	training job record
2 s3	iterations	30000	training job record
2 s3	batch size	10	training job record
2 s3	crop and patch size (z, y, x)	[17, 256, 256]	training job record
2 s3	learning rate	0.003	stage configuration
2 s3	keep rule	last_step (ckpt_030000.pth)	training job record
2 s3	validation every	2000	training job record
3 s4	iterations	20000	training job record
3 s4	batch size	10	training job record
3 s4	crop and patch size (z, y, x)	[17, 384, 384]	training job record
3 s4	learning rate	0.003	stage configuration
3 s4	checkpoint released	ckpt_020000.pth (last step)	training job record
3 s4	keep rule	last_step	training job record
all steps	scheduler	{"name": "diffusers_cosine_warmup"}	stage configuration
all steps	weight_decay	3e-05	stage configuration
all steps	seed	42	stage configuration
all steps	val_steps	32	stage configuration
all steps	label smoothing of the BCE loss	0.5	stage configuration
all steps	validation split: tile px	512	training job record
all steps	validation split: fraction of supervised pixels	0.05	training job record
all steps	validation split: seed	20261017	training job record
all steps	label set id	43965c9d120c	run register
1 s1	villa base config: optimizer	sgd	villa@3ea17f54a9b3:ink-detection/configs/aligned21_hybrid_3d2d.json, line 41
1 s1	villa base config: learning rate	0.01	villa@3ea17f54a9b3:ink-detection/configs/aligned21_hybrid_3d2d.json, line 42
1 s1	villa base config: batch size	64	villa@3ea17f54a9b3:ink-detection/configs/aligned21_hybrid_3d2d.json, line 48
1 s1	villa base config: patch size (z, y, x)	[17, 128, 128]	villa@3ea17f54a9b3:ink-detection/configs/aligned21_hybrid_3d2d.json, line 20
1 s1	villa base config: mixed precision	fp16	villa@3ea17f54a9b3:ink-detection/configs/aligned21_hybrid_3d2d.json, line 47
1 s1	resume source version (training record)	an earlier run	training job record
1 s1	resume stage	s1	training job record
1 s1	iteration the resume starts at	79001	training job record
depth warp	s1 stage block: enabled	true	training job record
depth warp	s1 stage block: max_offset	3	training job record
depth warp	s1 stage block: g_range	1	training job record
depth warp	s1 stage block: amp_max	2	training job record
depth warp	s1 stage block: sigmas_px	[8, 16, 32, 64]	training job record
depth warp	s1 stage block: void_probability	0.05	training job record
depth warp	s1 stage block: void_px	32	training job record
depth warp	s1 registration keys (registration record of the training chain)	warp_max_offset=3; warp_g_range=2; warp_amp_max=2	run register

step	setting	value	source
depth warp	s3 registration keys (registration record of the training chain), registered but not applied: this stage has no warp block	warp_max_offset=3; warp_g_range=2; warp_amp_max=2	run register
depth warp	s4 registration keys (registration record of the training chain), registered but not applied: this stage has no warp block	warp_max_offset=3; warp_g_range=2; warp_amp_max=2	run register
depth warp	s3 stage override carries no flat_z_depth_warp block	absent	training job record
depth warp	s4 stage override carries no flat_z_depth_warp block	absent	training job record
mad guard	relative threshold MAD_REL (scaled MAD under this fraction of the 1..99 percentile span)	0.01	training hook code
mad guard	lower bound of the scaled MAD (below it villa's own fallback already applies)	1e-6	training hook code
mad guard	UPSTREAM pull request proposal, NOT our hook: fallback whenever the scaled MAD is under this fraction of the value span	1e-2	upstream pull request proposal text

Appendix B
Graphics processors of the reads

The table of this appendix lists, for each model, how many of its fourteen unit reads ran on each type of graphics processor, as recorded by the pod of the read. Different types of graphics processor do not give bit-identical maps, and every comparison in this article is between maps from graphics processors, never between a map from a graphics processor and one from CPU.

model	4090	5090	A100 SXM	units
Rectina m1	14	0	0	14
Rectina m2	8	0	6	14
Reader v2	2	12	0	14
Reader v3	6	1	7	14

Appendix C
Provenance

Every number of the text is read from a table written by a tool. The scores of Rectina m1 against Reader v2: the AUC, the AUC on thin strokes and the false ink shares of both readers are read from one merge of the GPU reads, the cell file with the sha256 `d8a4fde07374d1d1cbde894efec73dd5b59c5467ecf8985450fd282e31eb9fe7`, and the paired tests from the statistics file with the sha256 `278491a321cd717ad85d9482b3b1d78ff1b8c9ae380c0d950285902b6737d953`, both pinned by checksum. The scores of Rectina m2 against Reader v3: the AUC, the AUC on thin strokes and the false ink shares of both readers are read from one merge of the GPU reads, the cell file with the sha256 `2723ada060fe999368565c3cc1c0258f5ae8a7ba1bc28e35e45d72ee704c1bfa`, and the paired tests from the statistics file with the sha256 `ae8ffe90e17ac6441f9ac77df57622f68dce07799de6d5da7438f97b712e9591`, both pinned by checksum. The scores of Rectina m1 against Reader v3: the AUC, the AUC on thin strokes and the false ink shares of both readers are read from one merge of the GPU reads, the cell file with the sha256 `2723ada060fe999368565c3cc1c0258f5ae8a7ba1bc28e35e45d72ee704c1bfa`, and the paired tests from the statistics file with the sha256 `ae8ffe90e17ac6441f9ac77df57622f68dce07799de6d5da7438f97b712e9591`, both pinned by checksum. The units and the energy blocks are in `held-out-units.csv` of the evidence of this article, the order of the units is that of the windows table of the bench in `five-model-bench/evidence`, the overlap of the controls with training is read from `control-overlap.csv`, the values of the recipe from the configuration files named in the table of appendix A, the control panels from the index named in the sections on the controls, and the states of the pull requests from the answer of the code host saved with its checksum at the build. The short codes of the held-out units in the tables stand for the following unit identifiers. 0009B·172639 is 20250510172639. 0009B·125754 is 20250919125754. 0009B·133208 is 20250919133208. 0009B·140325 is 20250919140325. 0500P2·055229 is 20250716055229. 0500P2·055230 is 20250716055230. 0343P·233748-0 is 20250904233748-0_b2. 0343P·172054-1 is 20250905172054-1_b2.

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About the author

Giovanni Pellerano is a computer engineer and whistleblowing hacktivist. Co-author and Project Lead of the free and open-source software, GlobaLeaks, he is responsible for software development and for the support of the community that has grown around it worldwide. He is a digital rights advocate currently learning more about mainstream generative AI development and how to resist against their various and profound harms.

He had not heard of the Vesuvius Challenge before Saturday 5 September 2026. Everything reported in this series was done after that date, which is stated because it bounds what the work can be assumed to rest on: no prior familiarity with the data, the tools, the community or the literature, just a curious and hyperfocused mind joining this new journey.

Declaration of generative AI in the research and writing process

During the preparation of this work the author used Claude Fable 5.1 and Claude Opus 5.5 (Anthropic) in order to carry out the research and to write the article: the models read the data, wrote and ran the tools, performed the measurements, and drafted the text. The author set the direction, supplied the domain judgement, challenged the conclusions, reviewed and edited the content, decided what was published, and takes full responsibility for the content of the publication.

The failure this arrangement is prone to is asserting a conclusion that no measurement supports. The comparison in this article is therefore fixed in advance as far as it can be: the rule for reading it is quoted from a declaration written before the scores existed, the units and the windows were fixed from

the labels before any map existed, and the table says where a measure cannot be taken instead of filling the cell.

What keeps the arrangement checkable is in the folder and not in this paragraph. Every number printed here expands a macro of `numbers.tex`, which a tool writes from the score files of the bench, and every table is written by the same tool: none is typed by hand, by the author or by the model.